Advanced Code Obfuscation

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The Simple Idea

If you have a **Secret Recipe**, and a thief steals it, they can use it instantly.

Obfuscation means rewriting the recipe in a secret code and mixing up the steps. It still makes the same food, but the thief can't understand *how*.

The 4 Layers of Protection

Layer 1
Changing Names (Lexical)

Layer 2
Mixing Logic (Flow)

Layer 3
Adding Traps (Opaque)

Layer 4
Mutation (Changing)

Each layer adds more confusion for the attacker.

Layer 1: Hiding Names

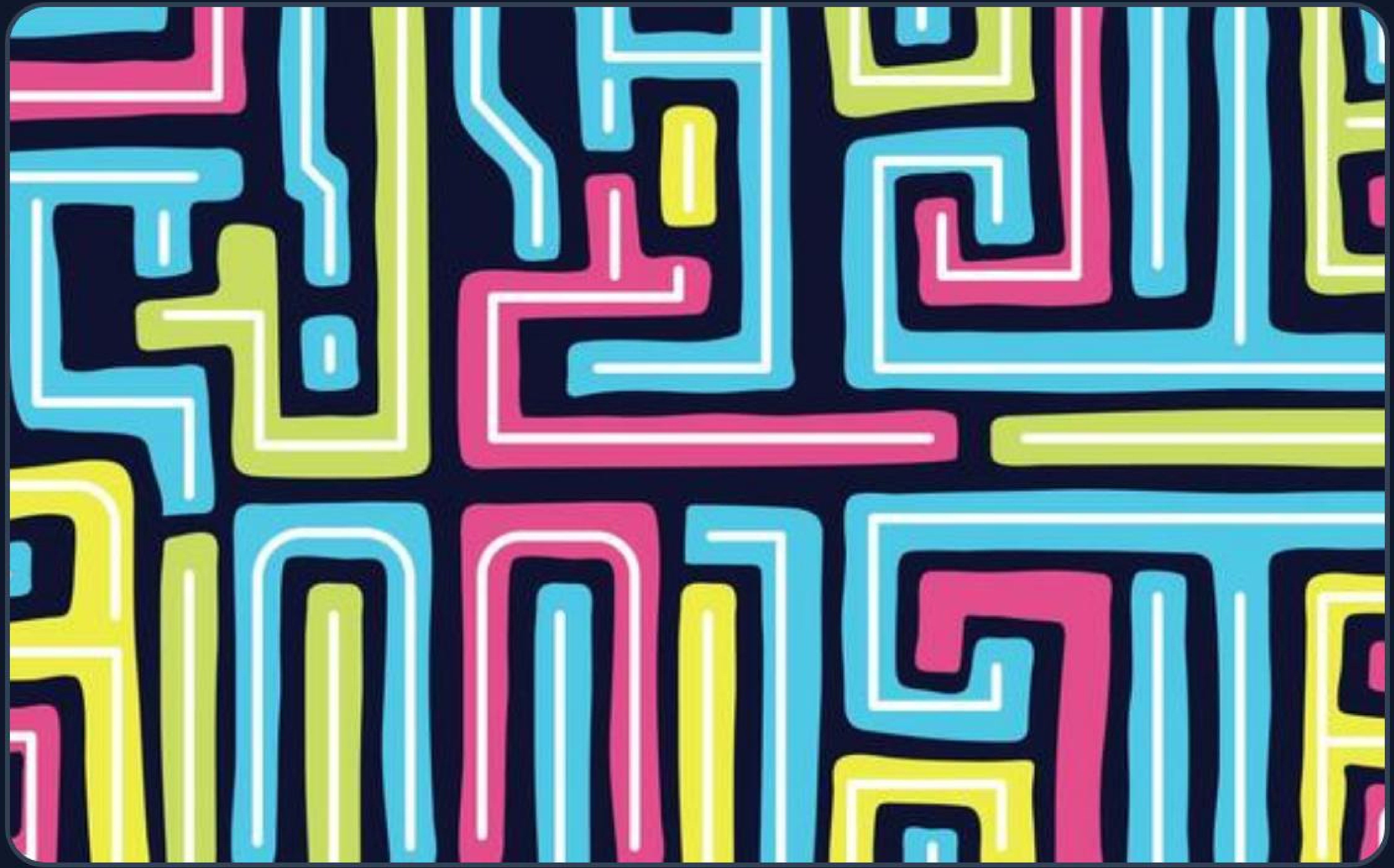
We replace helpful names like "password" with random letters.

This makes the code look like a foreign language to humans and hacking tools.

```
// Before
int user_password = 1234;

// After
int v_8b2d1c = 1234;
```

Layer 2: Breaking Logic



We break the code into small pieces and put them in a "Dispatcher".

Instead of going A to B, the code jumps around based on a secret "state".

```
while (state != 0) {
  switch (state) {
    case 1: do_math(); state = 4;
    case 4: print(); state = 0;
  }
}
```

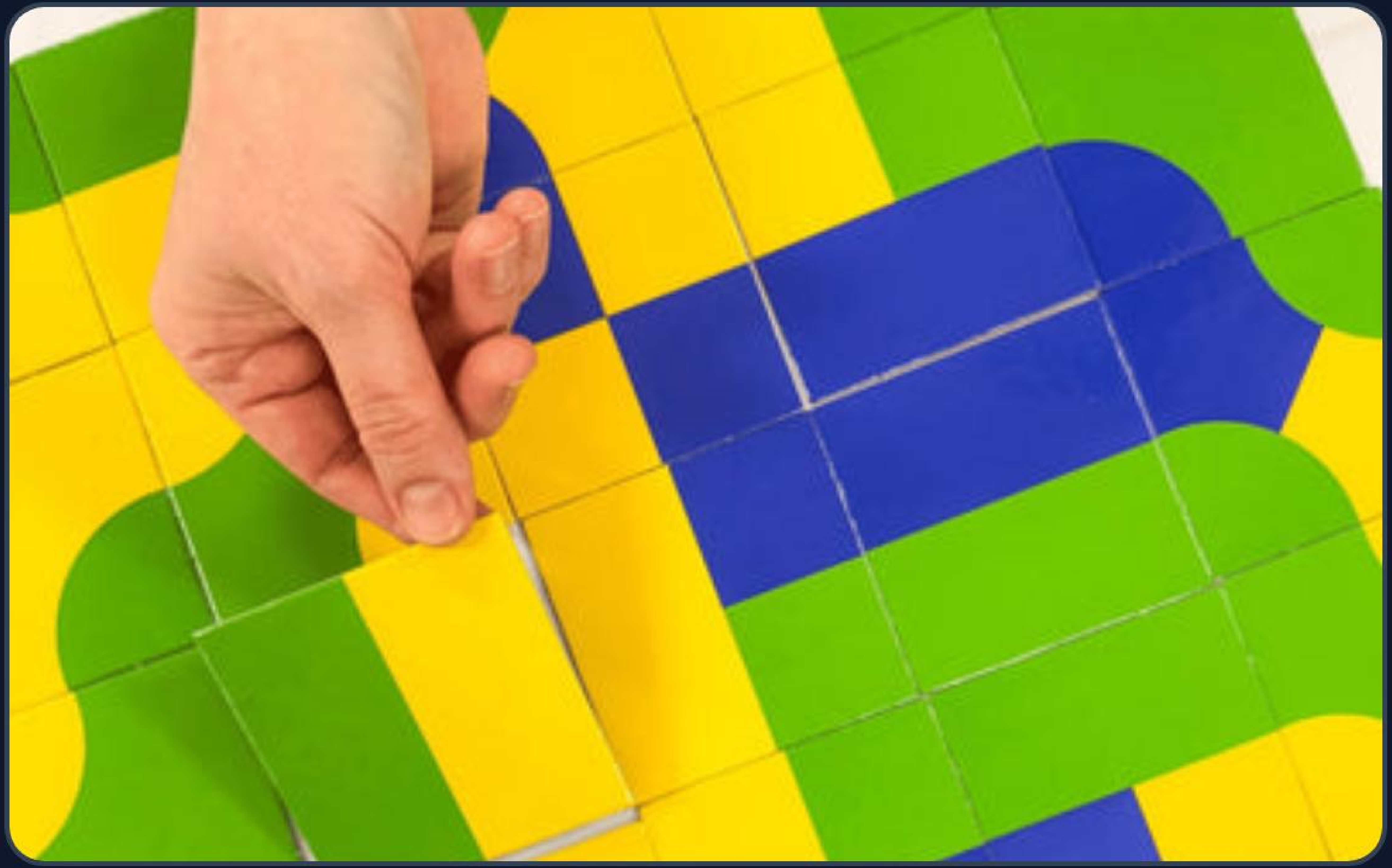
Layer 3: Adding Traps

We add fake code that can never be reached. We guard them with complex math.

Hackers waste time analyzing these dead-end paths.

$$(7 * x^2 - 1) \neq y^2$$

```
if (complex_math == true) {
  // Real code
} else {
  // 1000 lines of fake code
}
```



Layer 4: Code Mutation

The code changes its own shape while it is running!

It decrypts its instructions at the last second, so hackers can't find a pattern.

```
// Self-Modifying Code
asm volatile (
  "pop edx;"
  "xorb $0x28, (edx);"
  "call target;"
);
```

The Difference: Comparison

Fully Obfuscated

```
int f_x7(int a) {
  int s = 1;
  while(s != 0) {
    if((7*a-1)==2) { /* Trap */ }
    switch(s) {
      case 1: s = (a > 10) ? 2 : 3;
      case 2: return 1;
      case 3: return 0;
    }
  }
}
```

Original Code

```
int check(int x) {
  if (x > 10)
    return 1;
  else
    return 0;
}
```

The Final Result

10x

Higher Analytical Cost

By using these 4 layers, we make the software almost impossible to understand for hackers, ensuring our intellectual property is safe.

Static Analysis: FAILS | Dynamic Analysis: BLINDED

Questions?

Thank you for your attention.

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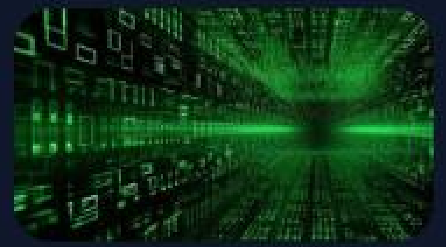
Image Sources



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